21 CFR PART 11 ELECTRONIC BATCH MANUFACTURING RECORD (eBHR)

1. BATCH EXECUTION & INLINE MULTIMODAL VISION TELEMETRY

- Facility: PLANT-XYZ-AUSTIN (Aseptic Bottling Line 03) | Batch Size: 24,000 Units of SKU-XYZ-204.
- At 10:14:22.418 UTC, GigE Camera #2 captured Frame FRAME-00841 showing an induction foil seal micro-tear.
- Gemini 3.8 Flash classified INDUCTION_FOIL_SEAL_TEAR (98.5% confidence) in 11.4ms; Pneumatic Gate DIVERT-01 fired in 14.2ms.

2. CLOSED-LOOP PLC PARAMETER CORRECTION & LOT RELEASE

- YieldMax correlated foil tear to induction coil voltage droop (-4.2V) and restored nominal 240.0V setpoint via OPC-UA.
- 23,999 conforming units passed hermetic seal verification and were released to Warehouse Slot A-01.

STRUCTURED ENTERPRISE LEDGER & VERIFICATION TABLE

Timestamp UTC	Event / Frame ID	Telemetry Reading	Automated Action	21 CFR 11 Sign-Off
10:14:22.406	FRAME-00841 Capture	4,200 Lux / 120 FPS	Sent to Gemini 3.8 Flash	SYS-VISION-03
10:14:22.418	Defect Classified	98.5% Conf (Foil Tear)	Trigger DIVERT-01 Gate	AGENT-VISIONINSPECT
10:14:22.432	Solenoid Actuated	14.2ms Gate Latency	Unit in Quarantine Bin 1	PLC-MODBUS-01
10:15:00.000	eBHR Deviation Log	VEEVA-DEV-2026-8841	Batch Released (99.99%)	QA-DIR-AUSTIN

DIGITAL CRYPTOGRAPHIC AUDIT SEAL & ENTERPRISE SIGN-OFF

Organization: XYZ Global Industries Corp. (XYZ-CORP-01) | ERP Sync: SAP S/4HANA + Cloud Spanner Twin

HMAC-SHA256 Verification Hash: 9f84c2a10e7b4d33a81e6c992b10d4f7e3a2118c6d004f9a | Status: APPROVED & VERIFIED

Authorized By: Elena Rostova (CSCO) & Vertex AI Agent Engine Autonomous Governance Enclave

4. CROSS-PILLAR AGENT HANDOFF & GOLDEN THREAD TRACEABILITY

- This document is automatically indexed in AlloyDB pgvector and linked to BigQuery dataset `supply_chain_twin`.
- Shared Enterprise SKUs referenced: SKU-XYZ-101, SKU-XYZ-204, SKU-XYZ-309, SKU-XYZ-412, SKU-XYZ-518, SKU-XYZ-625.
- Shared Facilities referenced: PLANT-XYZ-STUTTGART, PLANT-XYZ-AUSTIN, PLANT-XYZ-BENGALURU, RDC-XYZ-CHICAGO, FDC-XYZ-MUMBAI.

5. GOVERNANCE, MODEL ARMOR & DETERMINISTIC QUALITY GATES

- All agent write-backs require pre-execution `enforce_tool_contract()` validation and a unique `idempotency_key`.
- PII, vendor bank coordinates, and GSTIN/tax identifiers are tokenized via Google Cloud DLP prior to LLM context injection.

STRUCTURED ENTERPRISE LEDGER & VERIFICATION TABLE

Golden Thread ID	SKU ID	Purchase Order	Production Batch	Sales Order / Shipment
GT-XYZ-2026-001	SKU-XYZ-101	PO-XYZ-880142	BATCH-XYZ-STUT-904	SO-XYZ-20261022-801 / SHP-XYZ-90
GT-XYZ-2026-002	SKU-XYZ-204	PO-XYZ-880219	BATCH-XYZ-AUST-8841	SO-XYZ-20261022-802 / SHP-XYZ-90
GT-XYZ-2026-003	SKU-XYZ-412	PO-XYZ-880355	BATCH-XYZ-BLR-7712	SO-XYZ-20261022-803 / SHP-XYZ-91

DIGITAL CRYPTOGRAPHIC AUDIT SEAL & ENTERPRISE SIGN-OFF

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